

Emergency Treatment of Hyperthermia

Principles and Practice of Osteopathic Medicine I

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Heal.
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Session Objectives:

1. Summarize the mechanisms by which the body accomplishes heat loss
2. Define the mechanisms by which acclimation to hotter climates occur
3. Articulate the distinguishing factors of each level of heat-related illness
4. Describe the appropriate therapy for each level of heat-related illness
5. Identify the pitfalls of heat-related illness therapy

Heat Emergencies



Image from: <https://www.denofgeek.com/us/movies/top-37-classiest-satans-in-film-tv/88319/top-37-classiest-satans-in-films-or-tv-shows-ive-seen-in-the-past-two-weeks>

Epidemiology

- Varies with weather*
- Seen less in people who live in warmer climates than visitors**
 - Physiologic acclimatization
 - Cultural adaptation
- 688 heat-related deaths in the US in 2003
- 14,800 deaths in France during Heat Wave of 2010

*Journal of Duh

**Journal of Who Would Have Thunk

Mechanisms for Heat Loss

- Radiation - the transfer of heat by electromagnetic waves from a warmer object to a colder object
- Evaporation - heat loss by vaporization of water through:
 - sweat
 - respiration
- Convection - heat transfer by air or liquid moving across the surface of an object
- Conduction - heat exchange between two surfaces in direct contact e.g. ice packs

The primary methods used to cool hyperthermic patients are evaporation, convection and conduction

Methods of Heat Loss

- Behavioral control
- Sweating ($\uparrow$ cholinergic stimulation)
 - However, evaporation is limited in higher humidity. Therefore, sweating is less effective.
- Cutaneous Vasodilation through $\downarrow$ sympathetic tone
 - However, at higher ambient temperatures the body can no longer radiate heat. Therefore, vasodilatation is less effective.
- Conduction of heat
 - Into water is much more effective than into air
- Convection of heat
 - However, this too less ineffective above 32.2°C and 35% humidity. Therefore, fans alone are insufficient to cool patients in these conditions

Hemodynamic Response

- As a result of vasodilation, stroke volume decreases because of diminished preload
- To maintain cardiac output (CO), heart rate must increase
- Advanced age, dysrhythmias, negative chronotropic medications limit this tachycardic response so CO falls
- This may result in myocardial ischemia (in those with fixed coronary lesions), dysrhythmias or other end-organ decompensation.

Estimation of Water Loss

- Sweating begins as body temp rises above 37°C. Elderly do not begin sweating until 1.5°C higher body temperatures than do younger
- Every pound of water weight loss pound is 453ml, so replace with 500 ml (close enough)



Image from: <https://www.maxiaids.com/talking-bathroom-scale>

Medications that Interfere with Heat Removal

- Anticholinergic agents
 - Anticholinergic agents impair sweating
- Diuretics
 - Diuretics lead to volume depletion and decreased cardiac output
- Phenothiazines
 - Have anticholinergic properties; may lead to depletion of central stores of dopamine which interfere with the hypothalamic thermoregulatory center
- β -blockers and calcium channel blockers
 - Interfere with the appropriate cardiovascular response to heat
- Sympathomimetic agents
 - Cause cutaneous vasoconstriction, increased muscle activity and inhibition of sweating
- Ethanol
 - Inhibits secretion of antidiuretic hormone which leads to dehydration. Blunts heat-avoidance behavior.

Acclimation

- Lowers the hypothalamic thermal set point
 - triggers the onset of sweating at lower core temperatures
- Sweat production rate increases up to 3 liter/hr.
- ↑Aldosterone
 - ↑Plasma volume
 - Sodium conservation results in more efficient sweating, because the sweat is less salty
- Dilation of cutaneous blood vessels occurs at a lower core temperature



Image from:
<https://www.nydailynews.com/sports/baseball/yankees/ny-sports-yankees-notes-aroldis-chapman-20180810-story.html>

Milder Heat Illness

- Heat edema
 - 2° to cutaneous vasodilation and increased aldosterone → venous pooling → increased vascular permeability
 - usually just feet and ankles
 - Rx with elevation
- Prickly Heat – AKA *lichen tropicus*, *miliaria rubra*, or *heat rash*
 - Inflammation over clothed areas of blocked sweat glands
 - **Rx – light, clean, loose-fitting clothing; antihistamines; topical calamine, chlorhexidine or low-potency steroids**



Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline.
Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th Edition
www.accessmedicine.com
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Milder Heat Illness

- **Heat Cramps**

- Painful involuntary muscle contractions occurring during, or shortly after physical exertion in a hot environment
- Can occur in normal temperatures so it may be just the exertional component which causes them
- Exact cause is unknown but thought to be due to muscle fatigue, dehydration, and fluid and electrolyte disturbances
- Rx:
 - cool patient
 - replace fluid and electrolytes
 - Slow stretching of involved muscles
 - Diazepam, magnesium are unproven

Heat Stress AKA Heat Exhaustion – body temp is normal or slightly elevated ($<40^{\circ}\text{C}$) with normal mental status

- Water and/or salt depletion
- Headache; nausea; vomiting; malaise; muscle cramps; pre-syncope; dehydration; tachycardia; orthostatic changes; hemoconcentration; electrolyte findings depend on prior repletion efforts
- **Usually dehydrated**
- **Sweating remains intact despite volume depletion**
- **Elevate the legs**
- **If outside, move patient to a cool shady area**
- **Rx: replace water and electrolytes orally or intravenously**
- **Most often treated successfully outside a hospital**

Heat Stroke – body temp $>40^{\circ}\text{C}$ with altered mental status (AMS)

- Poorly understood mechanisms
 - Impaired or overwhelmed thermoregulatory response
- Life-threatening emergency
- Often has no sweating
- Hyperthermia → direct cell toxicity (denaturation of proteins and cell necrosis) → endotoxins → inflammatory cascade → vascular endothelial damage → multisystem organ failure
- Vascular permeability activates coagulation cascade which leads to Disseminated Intravascular Coagulation (DIC)

Classic Heat Stroke

- Occurs during periods of high ambient temperatures
- Usually in elderly or very young
- Occurs over a prolonged period
 - Volume and electrolyte disturbances are common
- Disproportionally affects those with psychologic, physiologic or pharmacologic impairment
 - e.g. Raynaud's disease; drugs such as anticholinergics, diuretics, antipsychotics, cocaine, ethanol
- Classic example – elderly patient who lives alone with no air conditioning

Exertional Heat Injury

- Usually with athletic activities
 - Usually in the first 4 days of athletic practice
- Less to do with high ambient heat and more to do with increased heat production
- Can span the gamut from milder symptoms to full-blown heat stroke



<https://www.pledgesports.org/2018/07/avoiding-heat-exhaustion/>

Confinement Hyperpyrexia

- Children left in cars
 - Non-ventilated cars can reach 60°C (140°F) in <10 minutes
 - Infants are at particular risk
 - Inchoate regulatory mechanisms
 - Cannot remove themselves
 - Increased heat gain because of increased body surface:volume
- Stowaways
 - In year 2000 nearly 500 (exact number is unknown) people died attempting to cross the US/Mexican Border



Image from: <https://www.newskarnataka.com/world/109-migrants-found-crammed-inside-mexican-truck>

Heat Stroke Central Nervous System Effects

- Mental status is the canary in the coal mine
- Cerebellum is particularly susceptible
- Irritability; confusion; bizarre behavior; hallucinations; **abnormal plantar responses (upgoing toes)**, decorticate and decerebrate posturing; hemiplegia and coma.
- Seizures are quite common and increase heat production just when you need it least



Image from:

<http://floda31.com/marije/works/canary-in-the-coal-mine/>

Organ Dysfunction

- Hypotension
 - Rx with IV fluids - **NOT ORAL because of altered mental status**
 - If refractory, *consider* inotropes (e.g. dobutamine or low-dose dopamine) but avoid those with α adrenergic effects (e.g. norepinephrine or dopamine) which cause cutaneous vasoconstriction
- Electrolyte disturbances
 - hypo or hyperkalemia
 - hypernatremia
- Renal Failure – usually the result of rhabdomyolysis
 - Tea-colored urine (not just dark from concentrated urine)
- Heart failure
- Liver failure
- Acute respiratory distress syndrome (ARDS)
- DIC



Image from: <https://en.wikipedia.org/wiki/Rhabdomyolysis>

Laboratory Studies to Consider

- **Stat Glucose**
- Complete Blood Count (here the excuse for needing a “baseline” study is real)
- Complete Metabolic Panel (CMP) – ↑K⁺ is common
- Arterial Blood Gas (ABG) – metabolic acidosis; respiratory alkalosis (both compensatory and as mechanism to dissipate heat)
- Coagulation Profile: International Normalized Ratio (INR); Fibrin Split Products (FSP) (to help dx DIC?)
- Creatine Phosphokinase (CPK) - rhabdomyolysis is common
- Electrocardiogram
- Lumbar puncture/CT of the head to evaluate other causes of altered mental status (AMS)

Treatment – immediate cooling and support of organ dysfunction

- Cooling methods
 - Evaporative
 - Immersion
 - Axillary and groin ice packs
 - Body bag with ice
 - Extracorporeal membrane oxygenation (ECMO) or Cardiopulmonary bypass
 - Cooling blankets
 - Cold water gastric, urinary bladder, rectal, or peritoneal lavage

Evaporative is most preferred method

- Cheap
- Readily available
- Effective
- **May cause shivering, which will raise body temperature—not good**
 - **Rx with benzodiazepines**
- Hard to keep monitor leads on
- Use a BIG fan



Image from: www.webrestaurantstore.com,
Jan. 2019



<https://www.tractorsupply.com/tsc/product/ventamatic-floor-drying-fan-yellow>

Cooling Method Notes

- All methods may be poorly tolerated, hinder resuscitative efforts or be impractical
- If one fails, then try another
- Goal is to **cool the CORE to normothermia (37°C)**
- **IV fluids** - little effect on cooling but are used to restore volume
- Avoid overshooting
- **Antipyretics such as acetaminophen and ibuprofen or other NSAIDs have no role in heat-related illness**
 - Because these interrupt the change in the hypothalamic set point caused by pyrogens such as seen with a fever. These medications are not expected to have any effect on a healthy hypothalamus that has simply been overloaded.

Disposition

- Healthy patients with heat edema, heat stress or, heat cramps may be discharged
- Heat stroke patients require admission – usually to the Intensive Care Unit

Sources:

- **Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 9e**, Judith E. Tintinalli, J. Stephan Stapczynski, O. John Ma, Donald M. Yealy, Garth D. Meckler, David M. Cline
- Zafran K. & Mecham C. (2018) Accidental hypothermia in adults, In J. Greyzel (Ed.), *UpToDate*. Retrieved January 18, 2018, from
<https://www.uptodate.com/contents/accidental-hypothermia-in-adults>
- Mecham C. (2018) Severe nonexertional hyperthermia (classic heat stroke) in adults, In J. Greyzel (Ed.), *UpToDate*. Retrieved January 18, 2018, from
<https://www.uptodate.com/contents/accidental-hypothermia-in-adults>
- [Poloyac M.](#), [Empey PE](#), Drug dosing during hypothermia: to adjust, or not to adjust, that is the question, [Pediatr Crit Care Med. 2013 Feb; 14\(2\): 228–229.](#)

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>